

2 Kernels and RKHS

1 Find the RKHS over $\mathbb{R}^d$ for each of the following kernels.

$$K(x, y) = x^\top y + c, \quad x, y \in \mathbb{R}^d, c \in \mathbb{R}^+.$$

Notice that $K(x_i, x_j)$ can be written as

$$K(x_i, x_j) = \langle \phi(x_i), \phi(x_j) \rangle \text{ with } \phi(x) = [x^\top, \sqrt{c}]^\top \in \mathbb{R}^{d+1}.$$

Then define H as

$$\begin{aligned} H = \left\{ f_v = \sum_i^n \alpha_i \phi(x_i) \mid f_v(x) = v^\top \phi(x) \right. \\ \left. \text{with } \langle f_v, f_w \rangle = \left\langle \sum_i^n \alpha_i \phi(x_i), \sum_j^m \beta_j \phi(x_j) \right\rangle = \langle v, w \rangle = \right. \\ \left. \sum_i^n \sum_j^m \alpha_i \beta_j \langle \phi(x_i), \phi(x_j) \rangle = \sum_i^n \sum_j^m \alpha_i \beta_j K(x_i, x_j) \right\} \end{aligned}$$

*I realize that n must equal m to make any sense.

H is a vector space since

$$\begin{aligned} f_{v+w} &= f_v + f_w \text{ since } f_{(v+w)}(x) = (v+w)^\top \phi(x) = (v^\top + w^\top) \phi(x) = f_v(x) + f_w(x) \\ f_{\alpha v} &= \alpha f_v \text{ since } f_{\alpha v}(x) = \alpha v^\top \phi(x) = \alpha f_v(x) \end{aligned}$$

H is complete since it over $\mathbb{R}^{d+1}$ which is of finite dimension.

$$K(x_i, \cdot) = \phi(x_i) \in H \text{ with } \alpha_k = \delta_{ki}$$

K has the reproducing property since

$$\langle f_v, K(x, \cdot) \rangle = v^\top \phi(x) = f_v(x)$$

Thus H is a RKHS.

$$K(x_i, x_j) = (x_i^\top x_j)^2 \quad x_i \in \mathbb{R}^d$$

Notice that $K = (x_i^\top x_j)^2 = (x_i^\top x_j) (x_i^\top x_j) = x_i^\top (x_j x_j^\top) x_i = x_i B x_i$.

Since B has the form

$$B = \begin{bmatrix} x_j^1 x_j^1 & x_j^1 x_j^2 & \cdots & x_j^1 x_j^d \\ \vdots & & & \vdots \\ x_j^d x_j^1 & x_j^d x_j^2 & \cdots & x_j^d x_j^d \end{bmatrix}$$

it is symmetric and thus $K(x_i, x_j) = x_i^\top S_B x_i$

Guess that $H = \left\{ f_S \mid f_S(x) = x^\top S x \text{ where } S \text{ is a } d \times d \text{ symmetric matrix} \right\}$

$$\text{with } \langle f_S, f_{S_B} \rangle = \sum_{nm}^d [S]_{nm} [S_B]_{nm}$$

H is a vector space since

$$\begin{aligned} f_{S_A} + f_{S_B} &= f_{S_A + S_B} \text{ since } x^\top S_A x + x^\top S_B x = x^\top (S_A + S_B) x \text{ and} \\ f_{\alpha S} &= \alpha f_S \text{ since } x^\top \alpha S x = \alpha x^\top S x \end{aligned}$$

H is complete since it is over a space of finite dimensions $d \times d$.

$K(x, \cdot) \in H$ since if we take $K(x_j, \cdot) = f_{S_B}$ then $K(x_i, x_j) = x_i^\top S_B x_j$

K has the reproducing property since each element of B is $x_j^m x_j^n$ we have

$$\langle f_S, K(x_j, \cdot) \rangle = \langle f_S, f_{S_B} \rangle = \sum_{nm}^d [S]_{nm} [S_B]_{nm} = \sum_{nm}^d x_j^n x_j^m [S]_{nm} = x_j^\top S x_j = f_S(x_j)$$

Hence H is a RKHS.